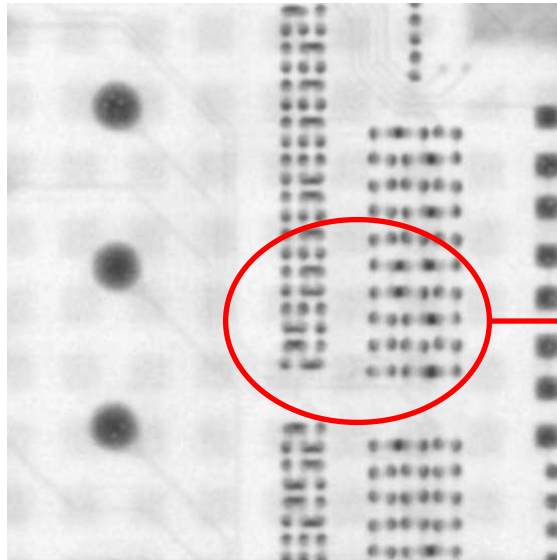


www.vitrox.com

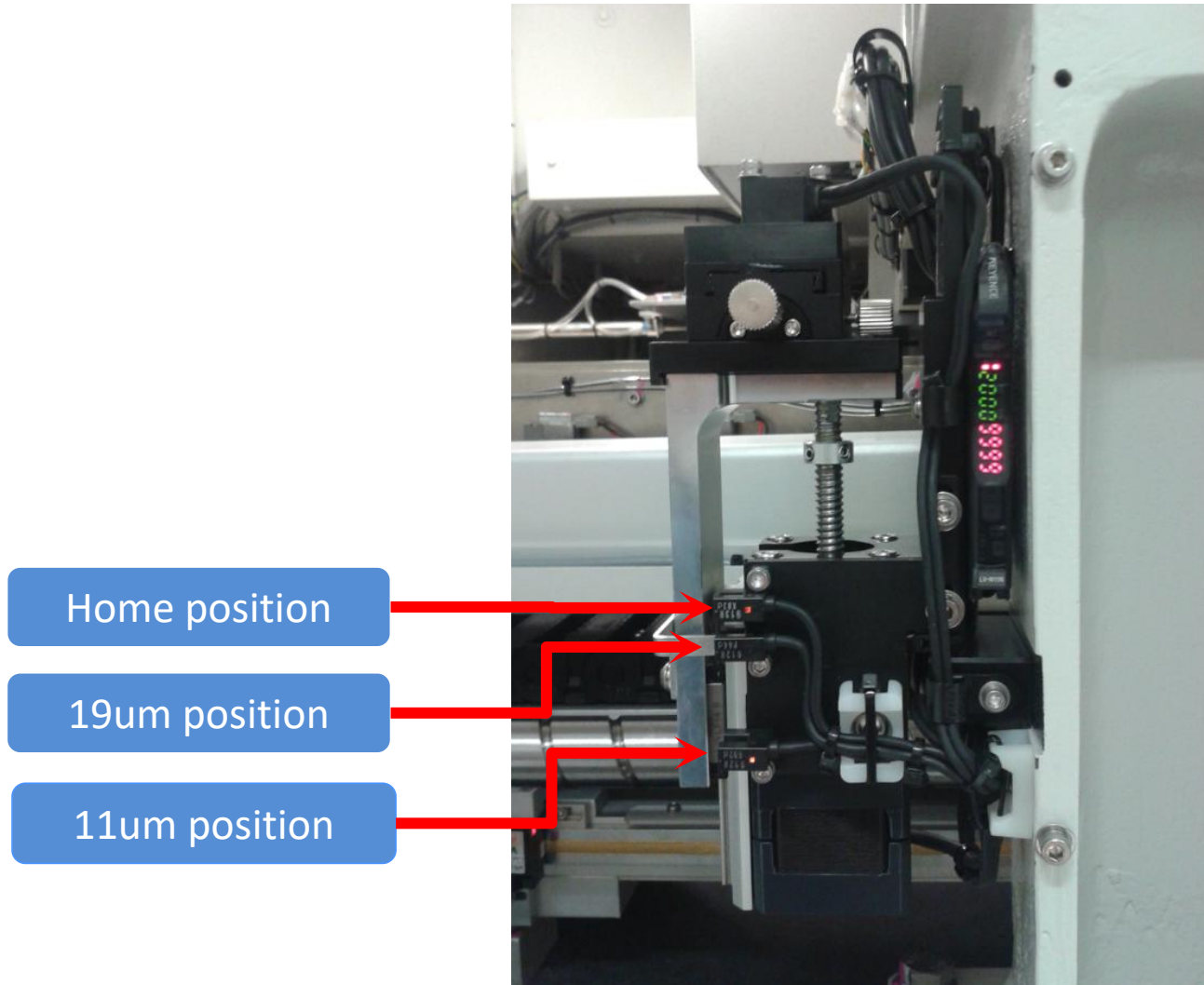
V810i S2EX Dual Magnification (Variable Mag) Hardware Overview



Low MagHigh Mag

- What is Magnification?
 - The ability of an optical (or x-ray) system to make an object appear larger.
 - Measured as the ratio of the size of the generated image to the size of the actual object
- Enhance the inspect-ability of V810i, with dual magnification which are Low (23 or 19 Micron) and High (11Micron) magnification.
- Only 1 low magnification and 1 High Magnification able to preset to run board inspection in same time

- Hmag function unable to disable for S2EX system type.
- When tube magnification change, tube position will become lower and more near to machine cameras array.
- Each magnification have difference allow load in board top clearance. Maximum component load into machine top clearance (calculated from **board** top surface) is
 - ❖ 50mm @ 23um resolution
 - ❖ 38mm @ 19um resolution
 - ❖ 11mm @ 11um resolution
- Machine equipped with Filter Height Sensor Module (Left and Right) to ensure loaded board is within define spec to avoid highest component crush with X-ray tube protrude portion.
- Board inspection will increase as XY stage movement scan path increasing and system need to run alignment point separately for low mag and high mag before start collect image.





Log Mag GUI Configuration

Automated Board Inspection

AXI System v810 - Main

System Type: v810 System S2EX (PSP+VM) 3262 S2EX
Low Magnification: M19
High Magnification: M11

Software Revision: 5.11.2 (11215) (Built 09/28/17 at 10:49:58)
Defect Packager Status: Running

Search Recipe/Variation:

Recipes Available on This System | Recipes Available in Recipe Database

No	Recipe name	Variation Name	Customer name	Version	Last modified	System Type
1	NEPCON		[not specified]	6.0	Nov 14, 2017 4:04:23 PM	S2EX

Change System Magnification

Warning
In order for the system's new magnification to take effect, the software will need to have a proper shutdown and run a full Confirmation, Diagnostic and Adjustment scheduled tasks after that.

Low Magnification: M19
High Magnification: M11

Restart Now Cancel

Selected Recipe Summary

Recipe name:
Customer Name:
Programmer Name:
Version:
Last modified:
System Type:
Number of boards:
Number of components:
Number of joints:
Number of joints tested:
Number of unstable joints:
Test coverage:

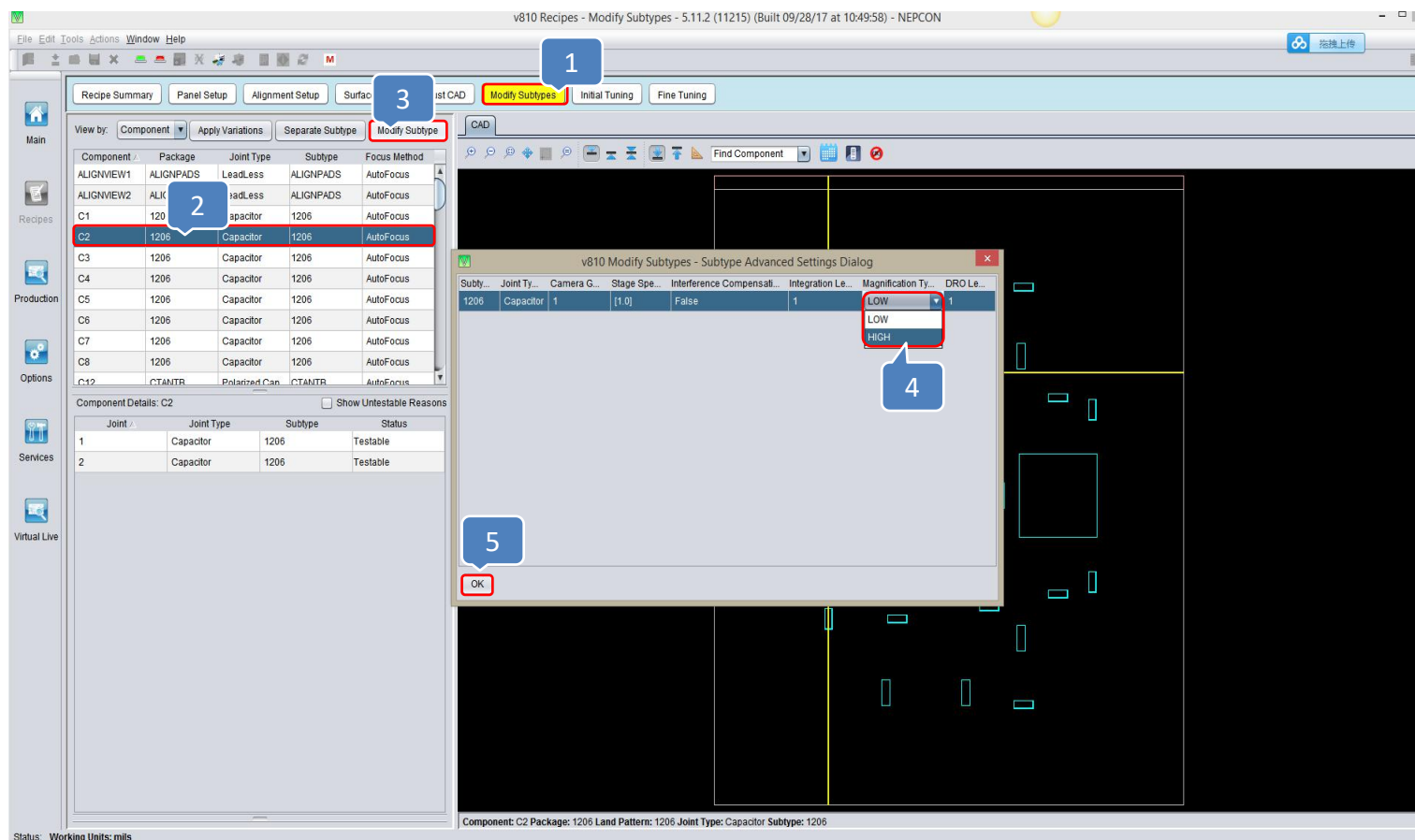
Status: There is 1 recipe available. Auto-pull recipes: OFF

- System only able to run with 1 low magnification (23um or 19um) and 1 High magnification (11um) in same time.
- Once magnification change, GUI will force restart to take effect



Automated Board Inspection

Hmag Inspection in Recipes Setting

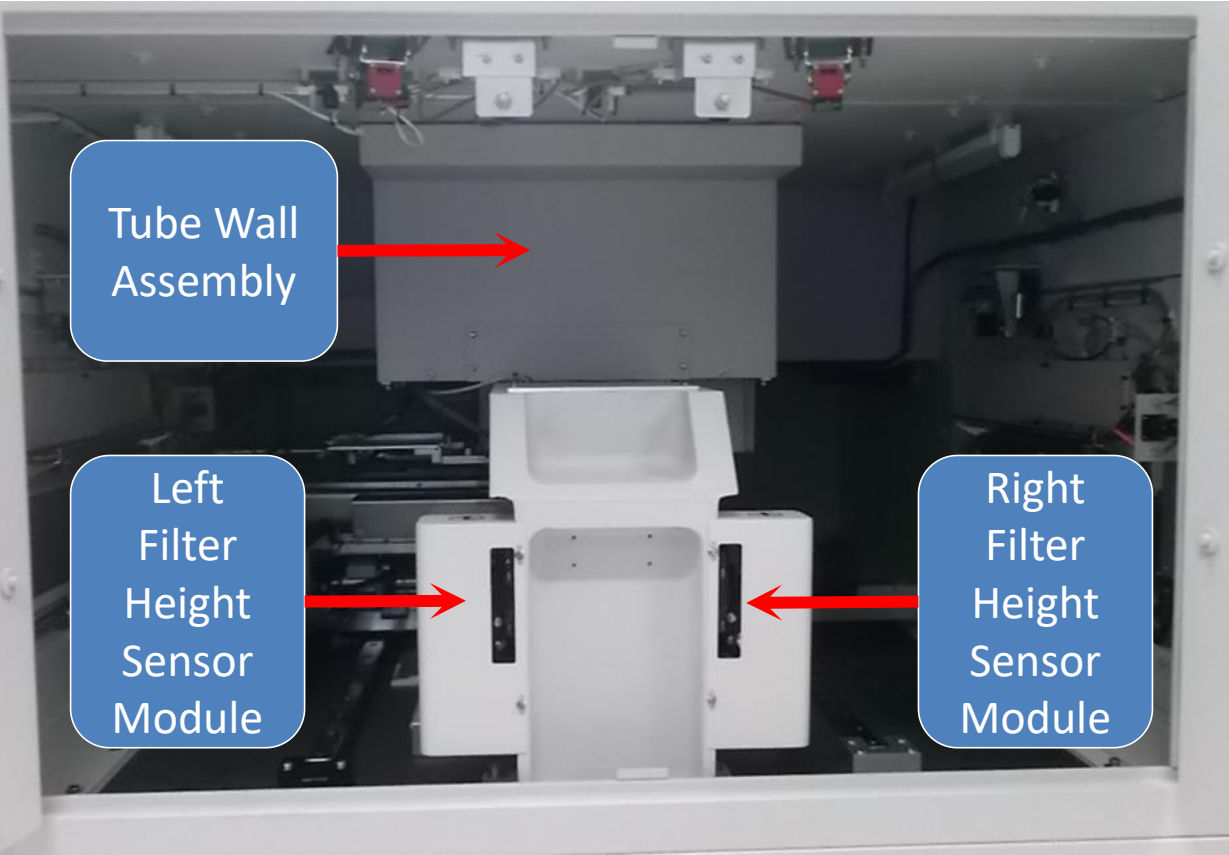


- Open the desire recipes, select the target component and go to modify subtype , change the “Magnification Type” to “High”, press “OK” and save.
- Recipes Alignment set-up need to be relearn as dual magnification used
- Inspection time will become longer as additional scan path added as Hmag inspection activated



Motorize Variable Magnification Hardware Overview

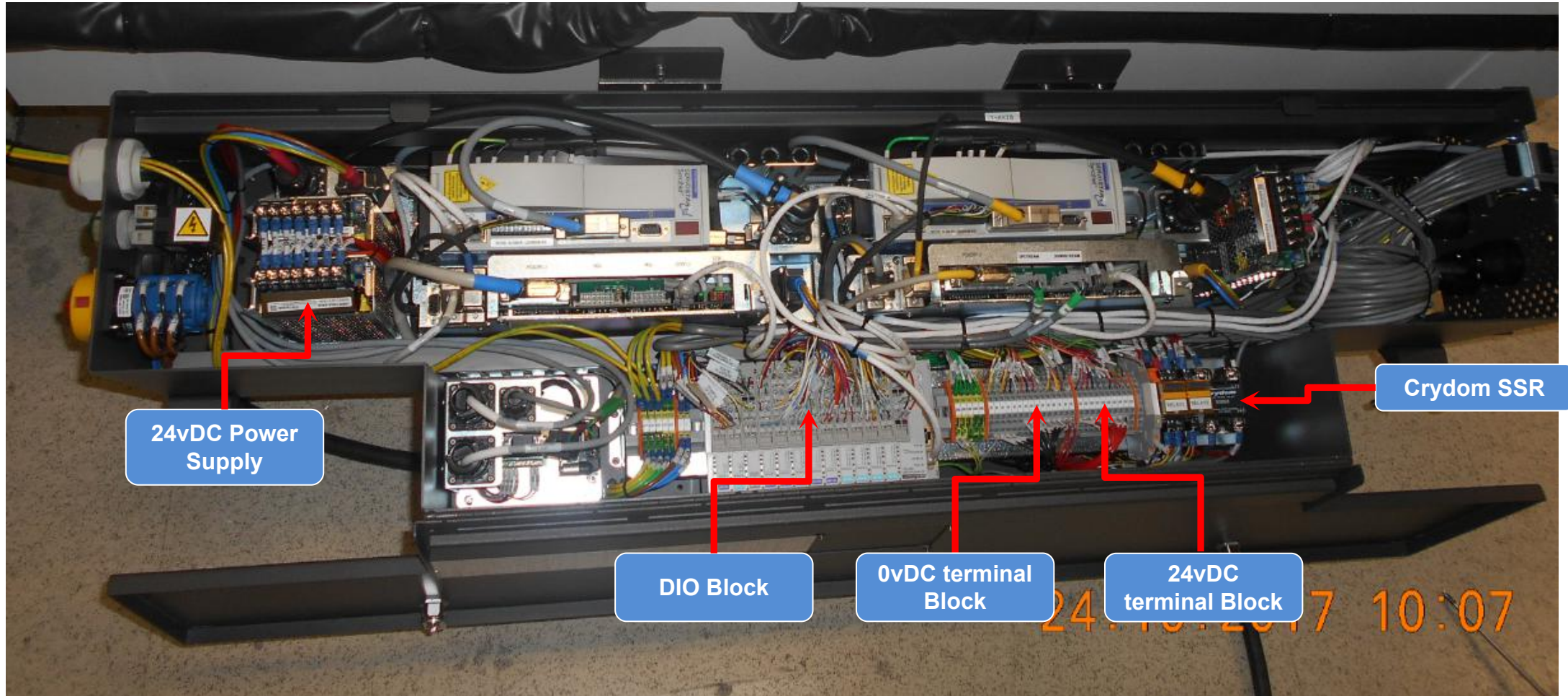
Automated Board Inspection

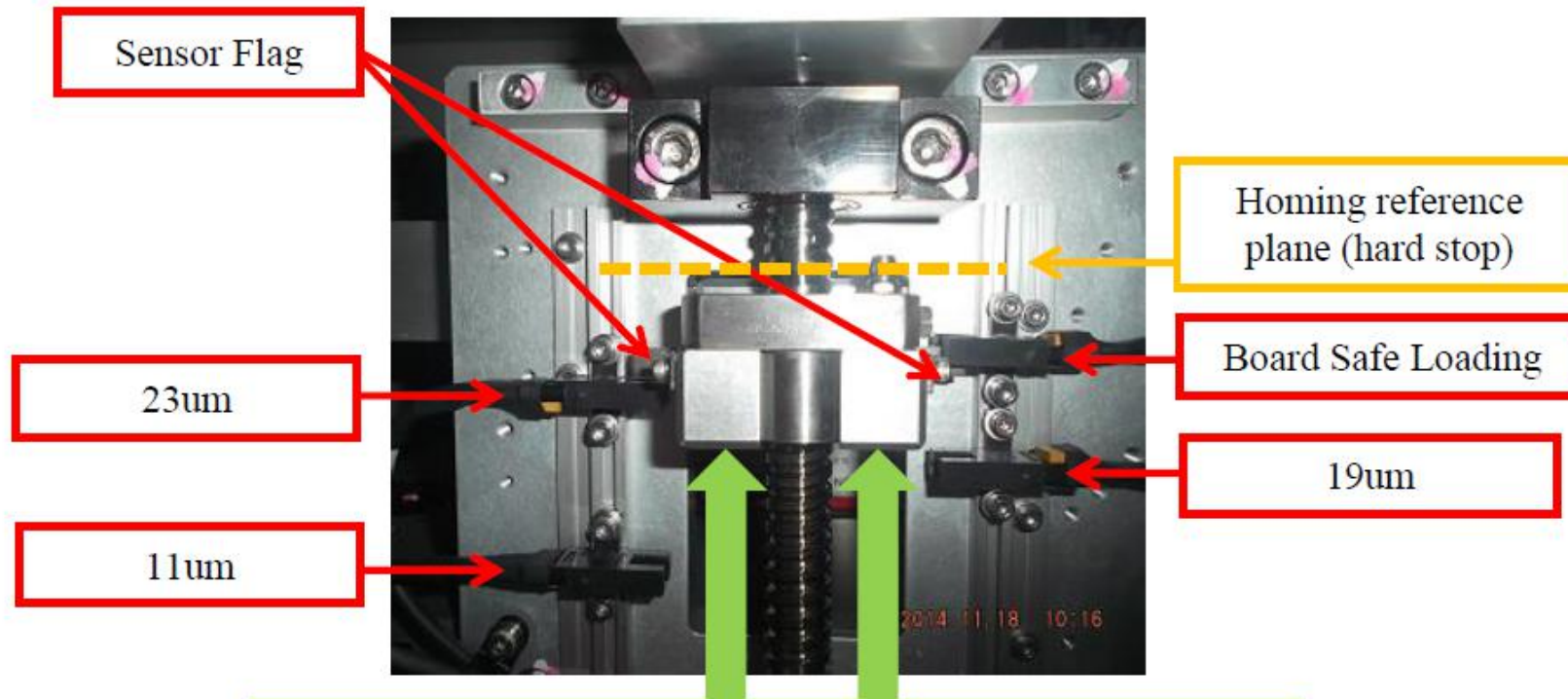


Front View



Side View





When enable the motor, the tube will move upwards and home to hard stop to establish the homing reference plane

v810 Service - Subsystem Status and Control - 5.11.2 (11215) (Built 09/28/17 at 10:49:58)

File Edit Service Tools Actions Window Help

Display Measurements in: Mils

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control X-ray Safety Test

Main Panel Handling Panel Positioner X-ray Source X-ray Camera Optical Camera Ethernet-Based Light Engine Light Tower SMEMA Interlocks Motion Control Digital IO Connected Processors Hardware Simulation Control

Show Hardware Configuration Description

Input Bits (Read Only)

- Left Outer Barrier Closed
- Left Panel Clear Sensor Blocked
- Optical Stop Controller Installed
- Interlock Switch ID Bit 4
- Interlock Switch ID Bit 0
- Z-Axis Servo On
- Panel From Upstream System Available
- Left Panel In Place Sensor Extended
- Optical PIP Sensor Reset On
- Z-Axis At High Mag Position 1 On
- X-ray Cameras Connected

Right Outer Barrier Closed

- Right Panel Clear Sensor Blocked
- Interlock Service Mode Off
- Interlock Switch ID Bit 3
- Inner Barrier Closed
- X-Ray Left Safety Level Sensor On
- Optical Stop Controller Done
- Left Panel In Place Sensor Retracted
- Panel Clamps Opened
- Motorized Height Level Sensor - Low Mag
- X-ray Cameras Powered On

Left Outer Barrier Open

- Right PIP Optical Stop Sensor Engaged
- Interlock Powered On
- Interlock Switch ID Bit 2
- Inner Barrier Open
- X-Ray Right Safety Level Sensor On
- Optical Stop Controller Error
- Right Panel In Place Sensor Extended
- Z-Axis At Low Mag Position 1 On
- Motorized Height Level Sensor - High Mag
- X-ray Cameras Power In Range

Right Outer Barrier Open

- Left PIP Optical Stop Sensor Engaged
- Interlock Switch ID Bit 5
- Interlock Switch ID Bit 1
- Z-Axis At Ready Safe Position On
- Downstream System Ready To Accept Panel
- Optical Stop Controller Enabled
- Right Panel In Place Sensor Retracted
- Z-Axis At Low Mag Position 2 On
- Motorized Height Level Sensor - No Alarm
- X-ray Cameras Phase Locked

Output Bits (Write)

Press button to change state

Left Outer Barrier Open

Right Outer Barrier Open

Enable Optical Stop Controller

Stage Belts On

Inner Barrier Closed

Light Stack Buzzer On

Left Panel In Place Sensor Extended

Stage Belts Direction Right To Left

Light Stack Red Light On

Z-Axis Enable Signal On

Panel From X-ray Tester Available

Motorized Height Level Sensor - Homing

Motorized Height Level Sensor - Position Bit2

Left Outer Barrier Open

Right Outer Barrier Open

Reset Optical PIP Sensor

Panel Clamps Opened

Light Stack Green Light On

Z-Axis Position Bit A On

Z-Axis Position Bit B On

Panel From X-ray Tester Not Tested

Motorized Height Level Sensor - Start Motion

X-ray Cameras Synthetic Trigger On Bar

Motorized Height Level Sensor - Position Bit0

Motorized Height Level Sensor - Position Bit1

Legend

ON OFF Error

ClearPath GUI Override

Tube 11um position

Tube 23um position

Tube 19um position

Tube safe load position

Override Switch button and back to safe load position

Move tube 23um position

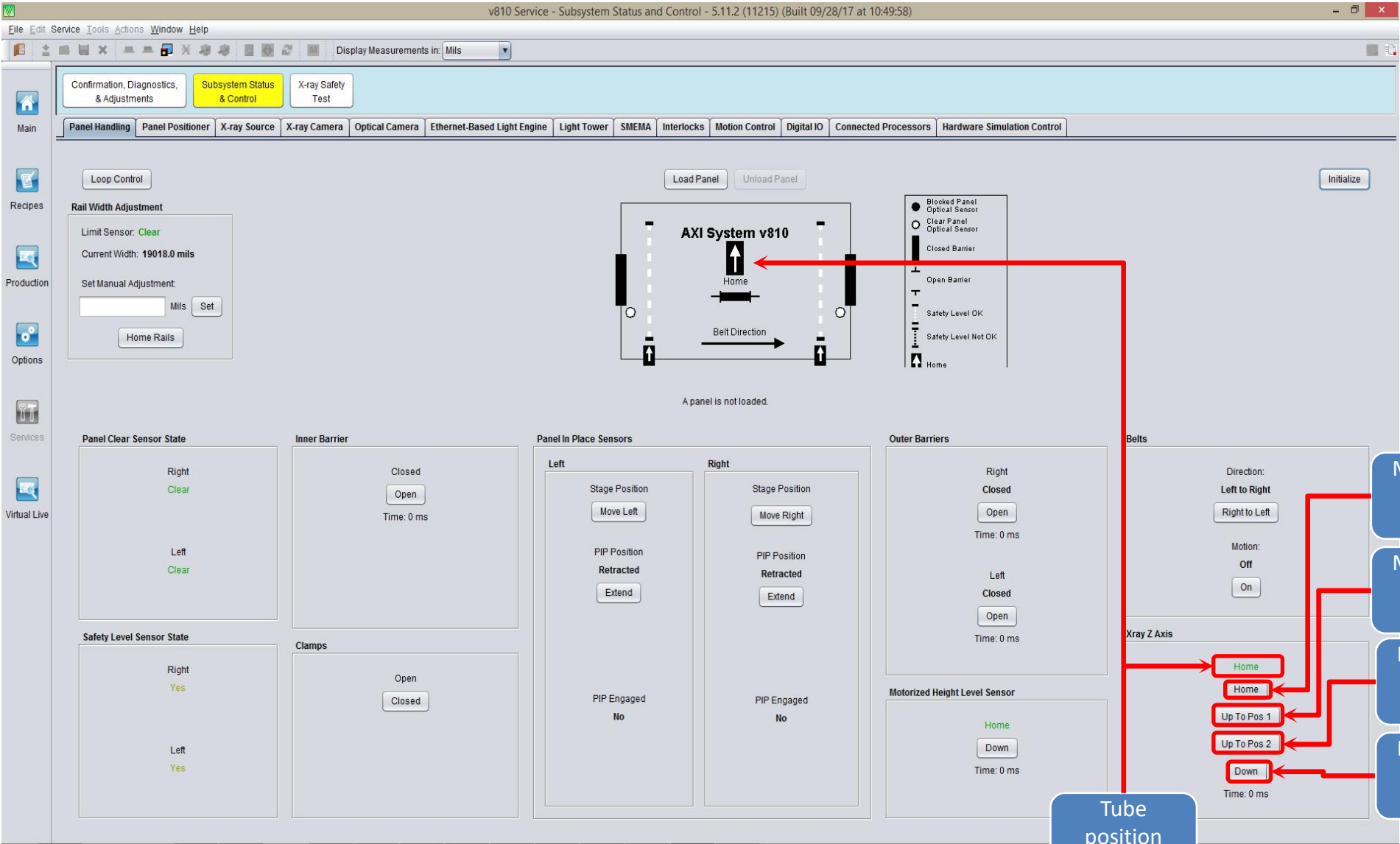
Move tube 11um position

Move tube 19um position



Variable Magnification – Panel Handling Interface

Automated Board Inspection



- Move tube safe load position
- Move tube 23um position
- Move tube 19um position
- Move tube 11um position

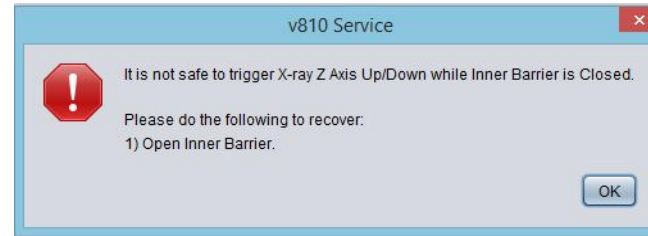
Tube position Status



Automated Board Inspection

I/O Function Checking

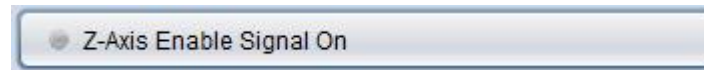
- Direct toggle either Input Bit A or Input Bit B or both without open inner barrier will prompt the error below.



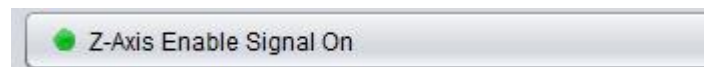
- Open Inner barrier in order to move the X-ray tube down.

Homing (Establish Reference position)

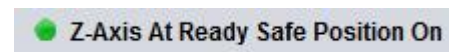
- Toggle the output bit “ Z-axis Enable Signal On” OFF.



- “Servo On” input bit will turn OFF. (* Clicking sounds can be heard as brake is activated to prevent X-ray tube movement when motor is disabled.
- Toggle the “ Z-axis Enable Signal On” ON.



- “Z-Axis At Ready Safe Position On” and “Z-axis Servo On” Input Bit will turn ON.





I/O Function Checking

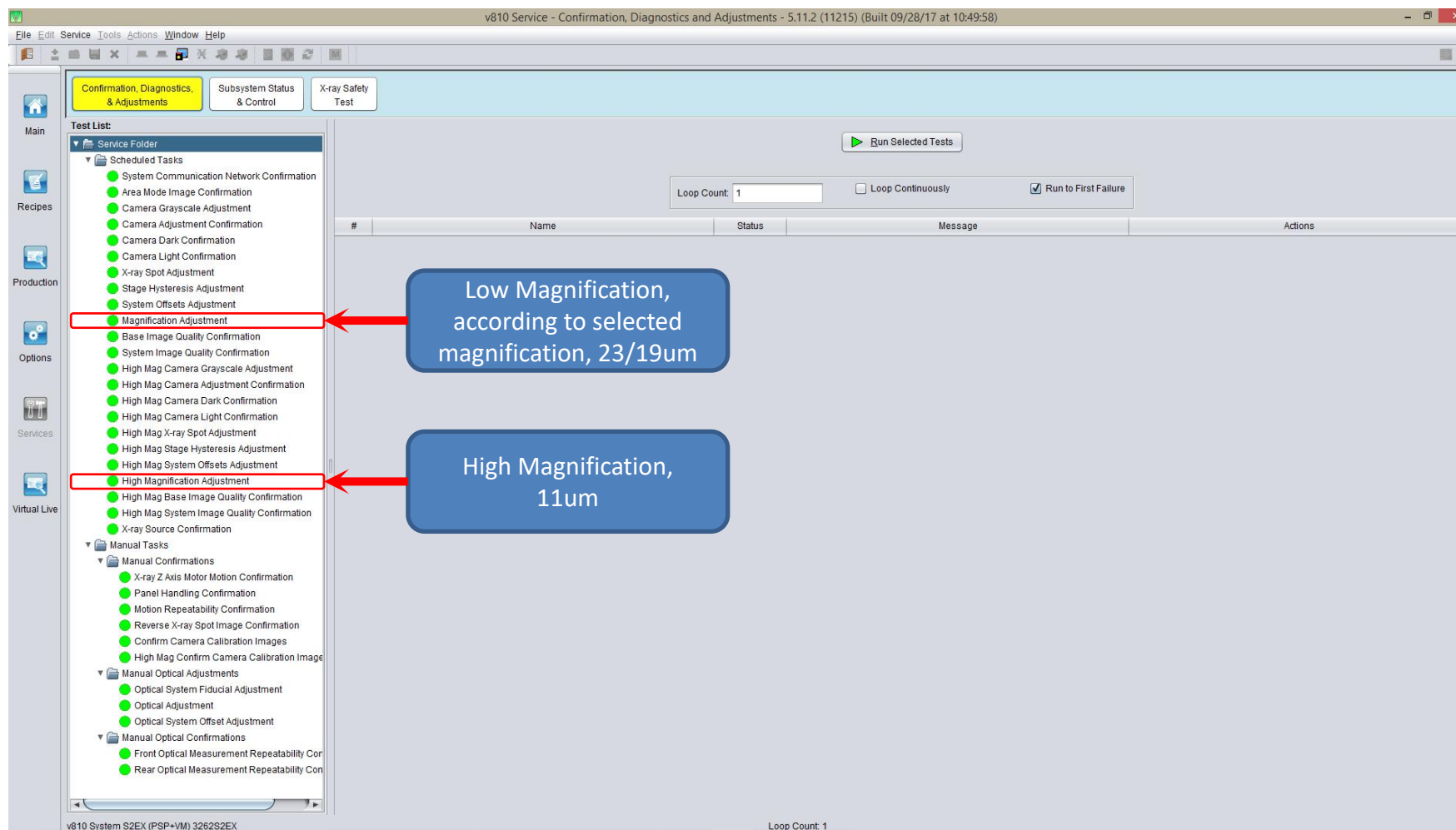
Automated Board Inspection

Position	Output Bit	Input Bit
	☐ Z-Axis Position Bit A On ☐ Z-Axis Position Bit B On	<u>Safe Board Loading Position</u> ☒ Z-Axis At Ready Safe Position On
	☒ Z-Axis Position Bit A On ☐ Z-Axis Position Bit B On	<u>23um Resolution</u> ☒ Z-Axis At Low Mag Position 1 On
	☐ Z-Axis Position Bit A On ☒ Z-Axis Position Bit B On	<u>19um Resolution</u> ☒ Z-Axis At Low Mag Position 2 On
	☒ Z-Axis Position Bit A On ☒ Z-Axis Position Bit B On	<u>11um Resolution</u> ☒ Z-Axis At High Mag Position 1 On

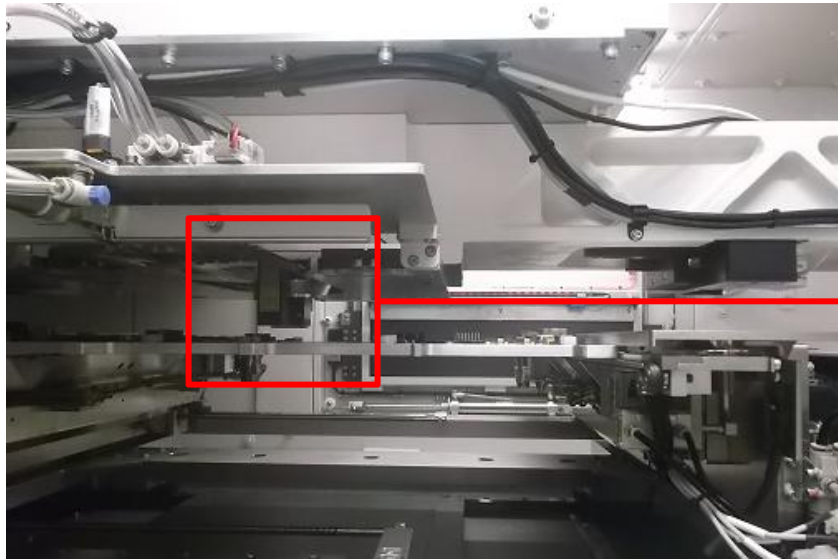


Automated Board Inspection

Confirmation, Diagnostic and Adjustment - Magnification task



- Log result directory, C:\Program Files\AXI System\v810\5.0\log\calib\Magnification
 - Low mag result log file name: magnificationYYYY.MM.DD.HH.MM.SS.log
 - High mag result log file name: highMagnificationYYYY.MM.DD.HH.MM.SS.log
- Sort by latest modify date



What is board top clearance?

- Available distance between X-ray tube and board top surface during conveyor in clamped condition.

Why the top clearance will vary?

- Different object plane resolution or magnification will cause the distance between X-ray and board is different.

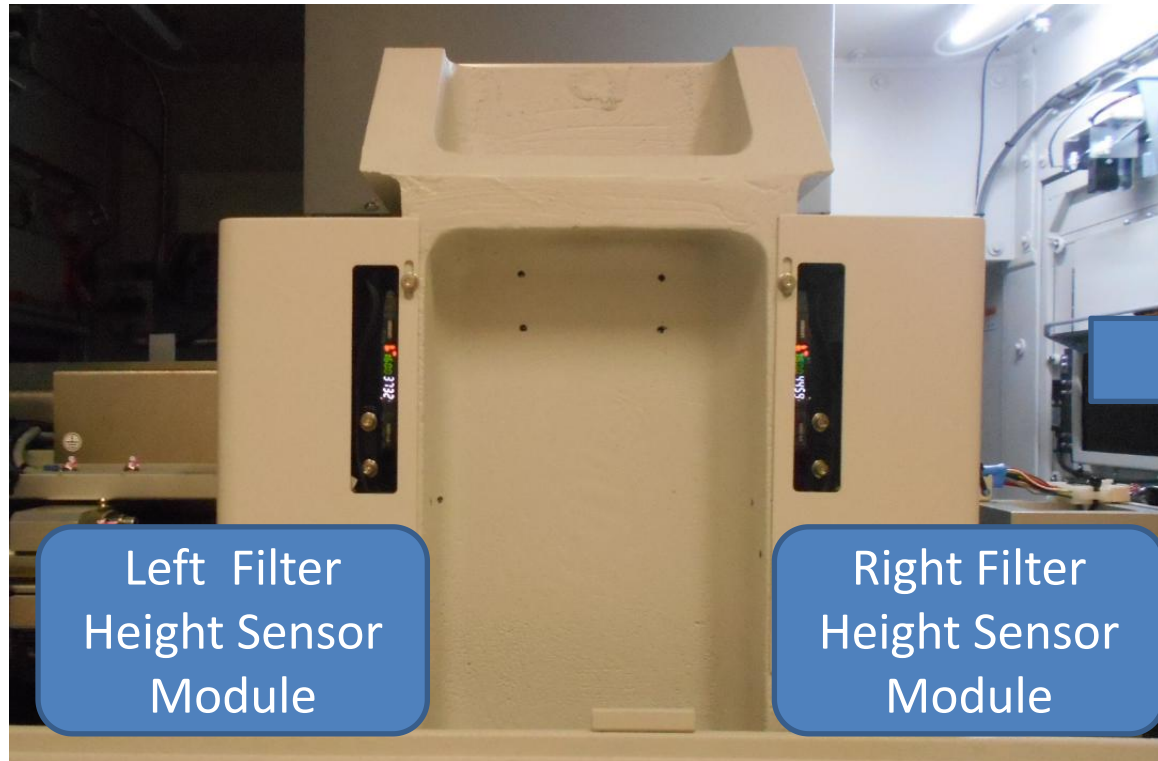
How the board top clearance varies with change of Magnification?

- Higher the magnification, smaller the board top clearance available.



Automated Board Inspection

Filter Height Sensor Module Hardware Overview



Left Filter
Height Sensor
Module

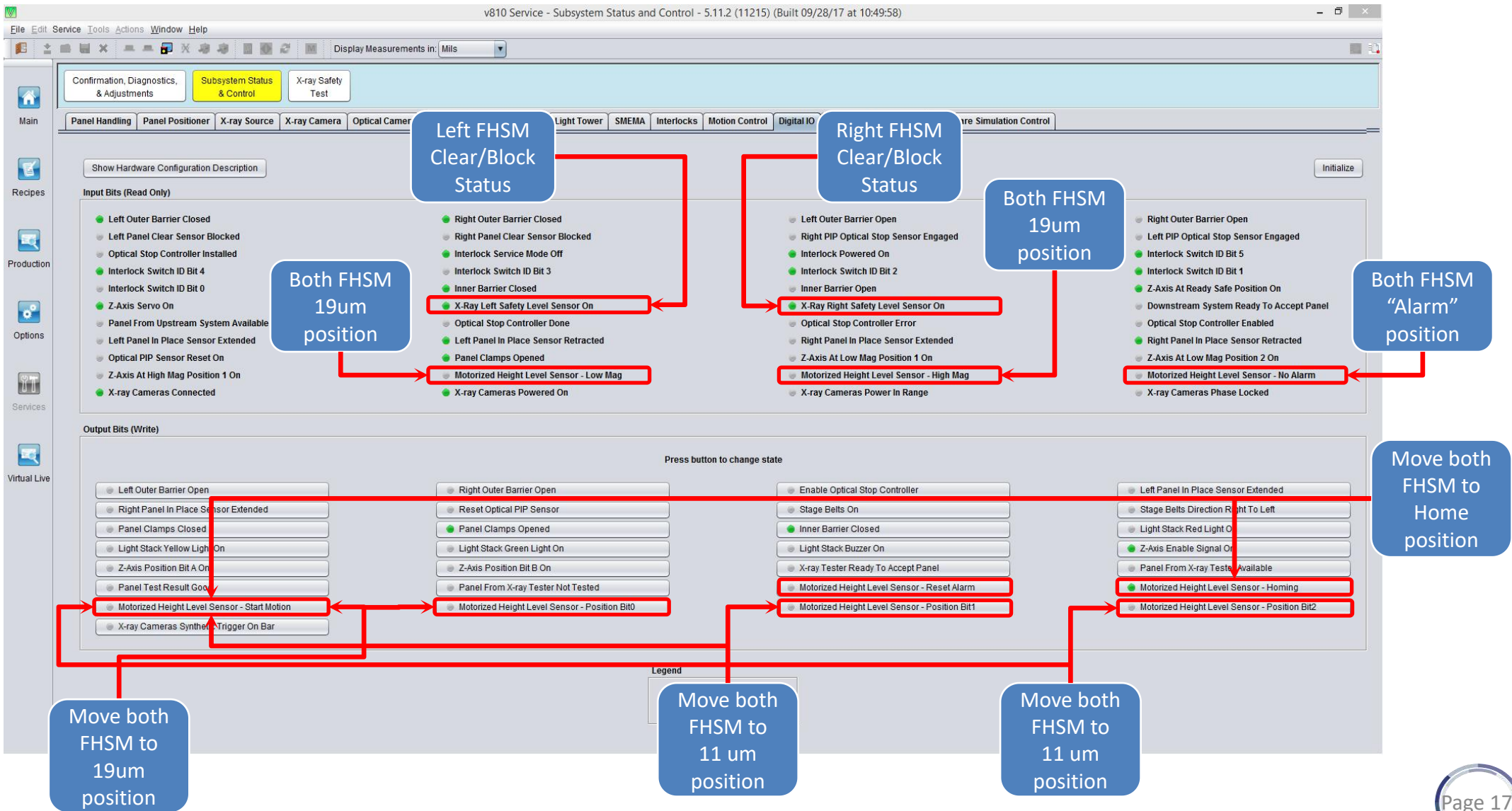
Right Filter
Height Sensor
Module





Filter Height Sensor Module – DIO Interface

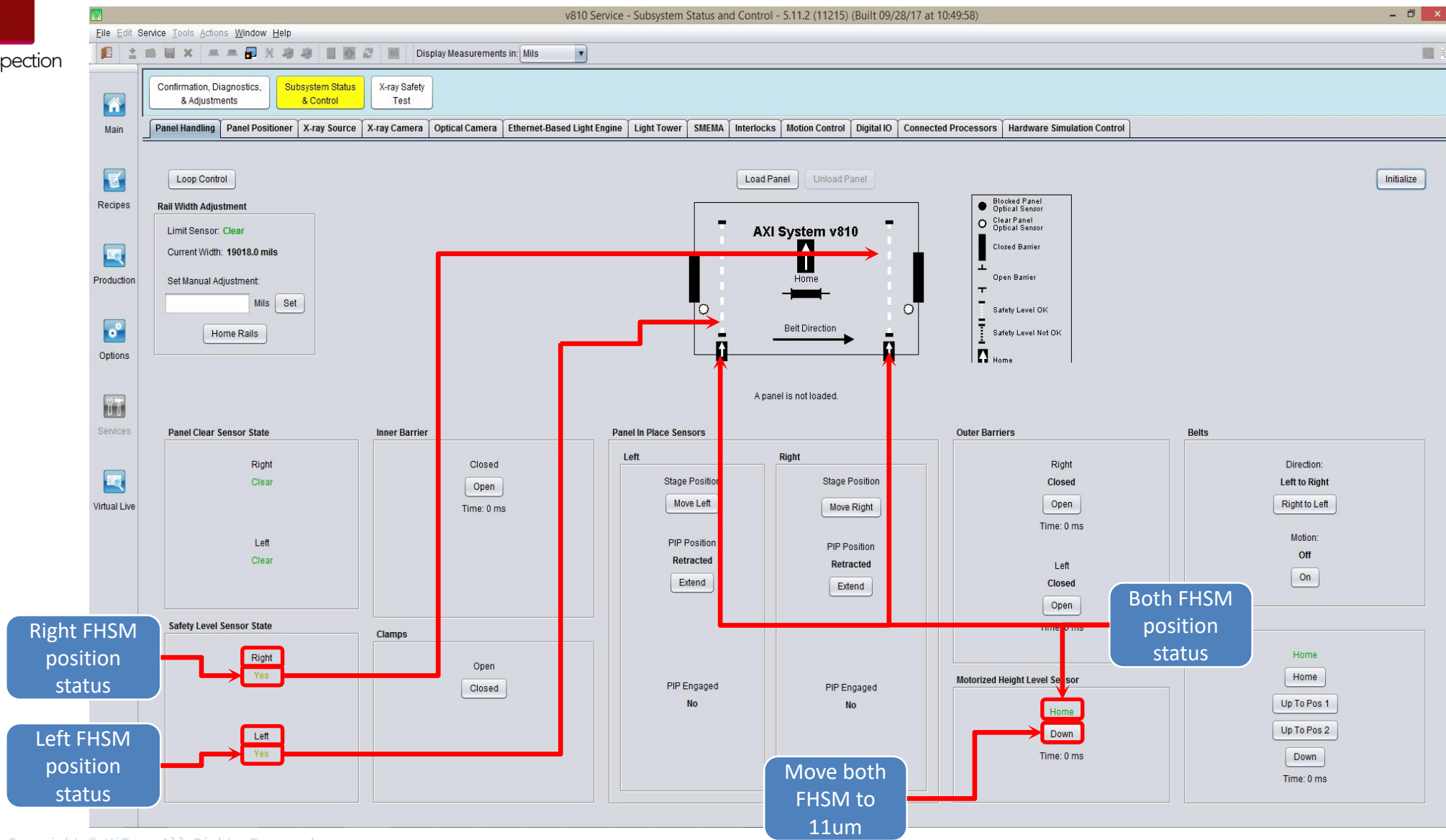
Automated Board Inspection





Filter Height Sensor Module – Panel Handling Interface

Automated Board Inspection





Automated Board Inspection

I/O Function Checking

- Toggle “ Motorized Height Level Sensor – Homing” to home the FHS modules

 Motorized Height Level Sensor - Homing

- If the homing is successful, the input bit “Motorized Height Level Sensor-No Alarm” should turn ON.

 Motorized Height Level Sensor - No Alarm

- If the homing is failed, input bit “Motorized Height Level Sensor – No Alarm” will turn OFF.
- Reset the Alarm by toggle “Motorized Height Level Sensor – Reset Alarm”

 Motorized Height Level Sensor - Reset Alarm

- FHS position setup (eg. 11um resolution).
 - Toggle “Motorized Height Level Sensor – Position Bit 0” to status ON.
Toggle “Motorized Height Level Sensor – Position Bit 1” to status OFF.
Toggle “Motorized Height Level Sensor – Position Bit 2” to status OFF.
 - Toggle “Motorized Height Level Sensor – Start Motion” to status ON.

 Motorized Height Level Sensor - Start Motion

- Input Bit “Motorized Height Level Sensor – High Mag” will turn ON.

 Motorized Height Level Sensor - High Mag



I/O Function Checking

Automated Board Inspection

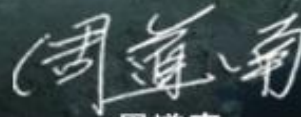
Position	Output Bit	Input Bit	Top clearance
	<u>N/A</u>	<u>23um Resolution</u> <u>N/A</u>	50mm
	☒ Motorized Height Level Sensor - Position Bit0 ☐ Motorized Height Level Sensor - Position Bit1 ☐ Motorized Height Level Sensor - Position Bit2	<u>19m Resolution</u> ☒ Motorized Height Level Sensor - Low Mag	38mm
	☐ Motorized Height Level Sensor - Position Bit0 ☒ Motorized Height Level Sensor - Position Bit1 ☐ Motorized Height Level Sensor - Position Bit2	<u>11um Resolution</u> ☒ Motorized Height Level Sensor - High Mag	11mm



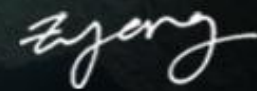
WEE KAH KHIM
Vice President & General Manager



GARY LEONG
Sales Director



周道南
Sales General Manager
(China & Taiwan)



SEOW ZI YANG
Business Development
Manager



FAHMI HELMI
Business Development
Manager